An endpoint correction to a printed positivity-rigidity conjecture

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25 October 2026

Abstract

The finite-sum rigidity statement in Conjecture 11 of Lóczi and Ketcheson constrains a coefficient that is absent from every hypothesis. We give a one-parameter family satisfying every premise and violating that endpoint conclusion, with a complete Lean certificate. This is a literal endpoint correction; it does not refute the intended rigidity statement restricted to the coefficients that occur in the sums.

1 The printed assertion

For $s \geq 1$, nonzero real q, positive real $\widetilde{q}$, and real coefficients $p_0, \ldots, p_{s+1}$, define

$$S_k = \sum_{m=0}^{\min(k,s)} \binom{s-1+k-m}{s-1} p_m q^m (-\widetilde{q})^{k-m}.$$

Conjecture 11 in [1], recorded as MathDB 325243, assumes $S_k \geq 0$ for $0 \leq k \leq s+1$. It concludes $p_0 \geq 0$ and

$$p_m = p_0 \binom{s}{m} (\widetilde{q}/q)^m \quad (1 \leq m \leq s+1),$$

and also $S_0 = p_0$ and $S_k = 0$ for $1 \leq k \leq s+1$. The upper coefficient index and the summation cutoff above agree with the original versioned source, not only with the database transcription.

2 Complete counterexample

Theorem 1. *For every nonzero real t, the choices*

$$s = 1, \quad q = \widetilde{q} = 1, \quad p_0 = p_1 = 1, \quad p_2 = t$$

satisfy all printed assumptions and violate the printed conclusion.

Proof. The parameter restrictions hold. Since every binomial coefficient in the three sums is $\binom{k-m}{0} = 1$, direct evaluation gives

$$S_0 = 1, \qquad S_1 = -1 + 1 = 0, \qquad S_2 = 1 - 1 = 0.$$

These are exactly all required inequalities, so every hypothesis holds. The claimed conclusion at $m = 2$ would be

$$t = p_2 = p_0 \binom{1}{2} (\widetilde{q}/q)^2 = 0,$$

contrary to the choice of t. □

More generally, each summand has $m \leq s$. Hence changing p_{s+1} never changes any S_k, for any k. The conclusion at that coefficient cannot follow without an additional condition.

3 Scope and formal certificate

The surrounding application of [1] takes p_m from derivatives of a polynomial whose degree is at most s, so its $(s+1)$st coefficient vanishes. The displayed free-coefficient conjecture does not impose this condition. It is plausible that the intended conclusion should stop at $m = s$, or that the endpoint coefficient was meant to be fixed at zero. This note makes no claim about the validity of either corrected formulation and is not a new substantive resolution of the intended rigidity problem. Its outcome is the exact correction to the printed statement.

The Lean file `formal/PositivityRigidity.lean` defines the exact sum and all hypotheses and conclusions. A total coefficient sequence merely supplies values outside the finite range; those values are never used. The theorem `sum_ignores_endpoint` proves the free-endpoint observation for arbitrary parameters. The remaining theorems prove every required inequality for the family above and refute the universal printed assertion in `printed_conjecture_false`.

All eight local declarations compile with Lean 4.34.1 against pinned mathlib dependencies. The extracted release contains the independent review, dependency pins and `verify_outcome.py`. Only the usual foundations `propext`, `Classical.choice`, and `Quot.sound` are permitted. The result does not depend on a numerical search, an admitted lemma, or assumptions imported from the conjecture.

References

[1] Lajos Lóczi and David I. Ketcheson. *Rational functions with maximal radius of absolute monotonicity*. arXiv:1303.6651v1, 2013. Section 2.5, Conjecture 11. https://arxiv.org/html/1303.6651v1.